

TECHNICAL ROADMAP



TEAM INFORMATION

Team Name	CIT-U: Shane-nanigans
Institution	Cebu Institute of Technology - University
Country	Philippines
Track	[✓] Climate Change
Team Leader Name	Lovely Shane P. Ong, onglvelyshane71@gmail.com, +63 9912227739

SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

Southeast Asia is the most disaster-prone region on earth. When the magnitude 6.9 earthquake struck Cebu on September 30, 2025, 79 lives were lost and 186,000 homes were damaged. What followed was not a failure of compassion; it was a failure of logistics intelligence. Dozens of NGOs operated with Google Forms and paper lists, no shared operational picture, and no way to find communities that had gone completely silent.

LUWAS is an AI-powered post-disaster logistics coordination platform built for NGO response coordinators across ASEAN. It answers three questions every coordinator faces: where do we go, what do we bring, and how do we get there, in Filipino, in real time, even without an internet connection.

LUWAS deploys a Silent Area detection engine for real-time community prioritization, TabPFN v2 transformer-based assistive supply forecasting, OR-Tools multi-team route optimization, and SEA-LION multilingual NLP parsing of Filipino and Bisaya field reports. Together these form an intelligent disaster coordination operating system where deterministic routing and scoring form the reliable core, and AI-assisted forecasting enhances every dispatch decision. Near-zero infrastructure cost at pilot and demo scale.

SECTION 2: TECHNICAL ARCHITECTURE

2.1 System Components:

Key Terms

Barangay: The smallest administrative unit in the Philippines, equivalent to a village or neighborhood. LUWAS operates at barangay resolution across the entire disaster-affected area.

Silent Area: A barangay that has received no confirmed volunteer contact within a critical time window, auto-prioritized by the system based on population size and hazard exposure

VRP (Vehicle Routing Problem): A class of combinatorial optimization problem that determines the most efficient set of routes for multiple vehicles serving multiple stops.

Inputs:

- **Coordinator inputs:** available teams, vehicle count, supply inventory, target barangay selection
- **Volunteer field reports:** structured app submissions and unstructured free-text in Filipino, Bisaya, or Tagalog
- **SMS shortcode submissions** from GSM-only field volunteers via Semaphore PH

- **Static datasets:** Project NOAH barangay hazard CSV (HDX), PSA 2020 Census, HDX Philippines pre-disaster vulnerability indicators, OpenRouteService Philippine road network
- **Real-time streams:** volunteer GPS positions and field report timestamps via Supabase Realtime WebSocket

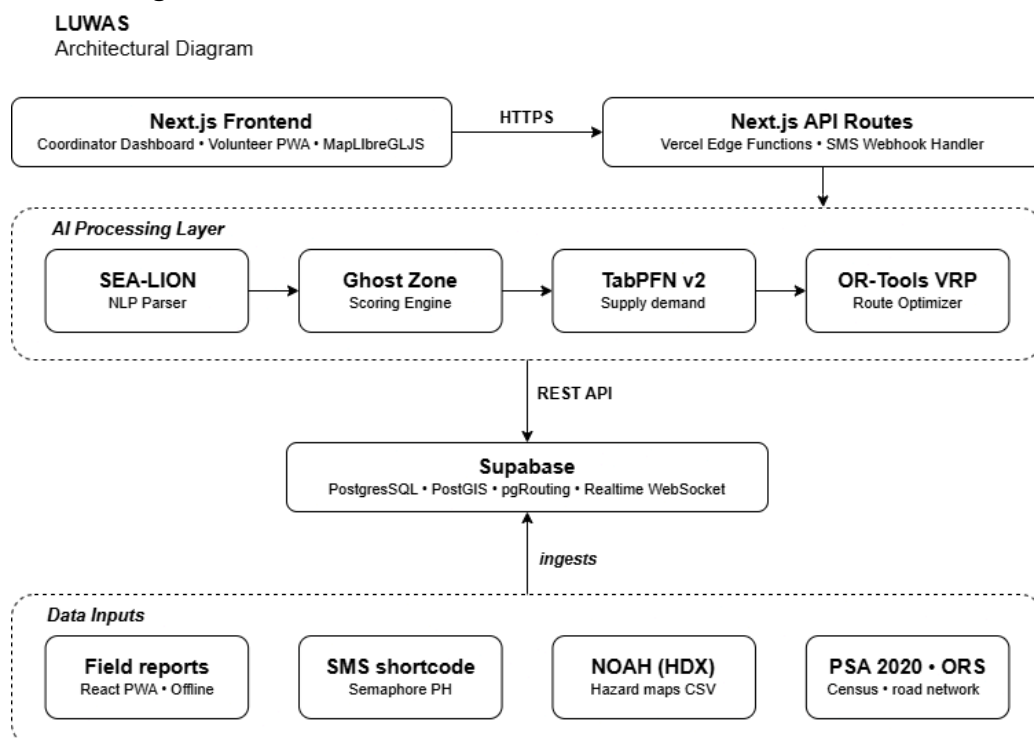
Processing Core:

- **Silent Area Scoring Engine:** Composite priority score per barangay computed every 15 minutes inside Supabase PostGIS, weighted by population density, Project NOAH hazard exposure, and time-decay since last confirmed field report. Fully deterministic, transparent, and explainable. This is the reliable operational core of LUWAS.
- **TabPFN v2 Assistive Supply Forecaster:** Transformer-based tabular foundation model (Prior-Labs/TabPFN-v2-reg) served via FastAPI on Hugging Face Spaces. Performs in-context learning on HDX historical disaster supply records to generate per-barangay food, water, and medical supply quantity estimates pre-dispatch. Positioned as an assistive forecasting layer: coordinator retains full override authority. No cold-start. Ground truth labels generated automatically by incoming field reports, enabling progressive accuracy improvement as deployment data accumulates.
- **OR-Tools VRP Solver:** Google's Vehicle Routing Problem metaheuristic solver consumes TabPFN v2 supply predictions as cargo manifests, Silent Area scores as stop weights, and a pgRouting real-road distance matrix. Produces optimized multi-team routes in milliseconds.
- **SEA-LION NLP Parser (via AI Singapore public API):** Receives unstructured Filipino, Bisaya, or Tagalog field reports and performs semi-automated structured extraction: location, population estimate, needs severity, road status. Low-confidence extractions are flagged for coordinator review before database insertion rather than auto-committed. Output normalized into Supabase identically to app-submitted reports. Gemini 2.5 Flash serves as high-availability fallback.
- **pgRouting Graph Traversal:** Road segments flagged impassable by volunteers trigger dynamic edge weight updates inside Supabase PostgreSQL, propagating to all subsequent OR-Tools routing computations with zero external API dependency.

Outputs:

- **Live MapLibre GL JS operational map** with three-state Silent Area pins: white (reached), amber (escalating), red (critical)
- **Optimized per-team dispatch routes** with ETAs and TabPFN v2 supply manifests per vehicle
- **Structured field records** from app submissions and SEA-LION-parsed free-text
- **SMS-triggered map pins** within 30 seconds of GSM-only volunteer shortcode submission

2.2 Architecture Diagram:



SECTION 3: AI APPROACH & MODEL SELECTION

- **Primary AI Approach:** [✓] Machine Learning [✓] NLP [] Computer Vision [✓] Generative AI [] Other: _____
- **Model Selection:**

TabPFN v2 (Prior-Labs/TabPFN-v2-reg) | Assistive Supply Forecasting

Transformer-based tabular foundation model purpose-built for small, mixed-type, sparse datasets. In-context learning treats HDX historical disaster supply records as the context window, delivering supply quantity estimates from Day Zero with no training pipeline required. Published benchmarks confirm TabPFN v2 outperforms CatBoost, XGBoost, and LightGBM on datasets under 10,000 rows, matching LUWAS's barangay-scale feature matrix.

Positioned as an assistive layer: coordinator reviews and overrides all forecasts before dispatch confirmation. Validated metric: MAE between predicted and coordinator-confirmed supply quantities, benchmarked against the equal-distribution baseline. IFRC logistics research documents that uncoordinated equal distribution wastes 25 to 40% of available supplies. As labeled field report data accumulates across deployments, the prototype path scales to a fine-tuned domain-specific regression model, moving beyond in-context learning toward a persistent humanitarian logistics forecasting model.

SEA-LION (AI Singapore) | Multilingual NLP Parsing

Called via AI Singapore's public inference API. The only production-accessible LLM trained on Southeast Asian corpora including Filipino, Bisaya, and Tagalog. Deployed as a semi-automated structured extraction engine: unstructured volunteer reports are parsed into normalized database records with confidence scoring. Extractions below the confidence threshold surface to the coordinator for manual field completion before committing to Supabase. This positions SEA-LION honestly as a high-value assistive parser, not a deterministic transformation engine, while still eliminating the majority of manual transcription burden from field volunteers.

OR-Tools VRP (Google) | Multi-Team Route Optimization

Graduate-level metaheuristic combinatorial optimization producing near-optimal multi-team humanitarian dispatch routes in milliseconds. Simultaneously consumes a real-road pgRouting distance matrix, Silent Area priority scores, and TabPFN v2 cargo manifests. The primary deterministic AI output of LUWAS: fast, reliable, and explainable.

SECTION 4: DATA STRATEGY & ETHICS

Data Sources:

Dataset	Source	License	Role
Project NOAH Hazard Maps	HDX	Open Government Data	Barangay flood, landslide, storm surge exposure
Pre-Disaster Indicators	HDX OCHA Philippines	CC Attribution	Housing type, water access, evacuation center count
PSA 2020 Census	Philippine Statistics Authority	Open Government Data	Population density per barangay
Historical Disaster Records	HDX OCHA	CC Attribution	TabPFN v2 in-context supply training examples

Road Network	OpenRouteService	ODbL	pgRouting graph seed data
Barangay Boundaries	OpenStreetMap	ODbL	Spatial administrative reference

Zero proprietary datasets. Zero web scraping. Zero synthetic data. All sources explicitly permit research and development use and are available through manual downloading / API.

Data Quality and Cleaning: All static datasets are government or UN-validated. Incoming field reports are PostGIS reverse-geocoded against barangay boundaries before insertion. SMS submissions are deduplicated via device fingerprint clustering. Anomalous supply values are flagged for coordinator review before TabPFN v2 ingestion.

Bias Mitigation and Fairness: Silent Area scoring applies identical population-weighted criteria across all barangays regardless of socioeconomic tier or urban/rural classification, directly countering systematic under-prioritization of geographically isolated communities documented in Philippine disaster response literature. SEA-LION multilingual parsing eliminates English-language bias from field data collection. TabPFN v2 outputs are monitored via segmented MAE analysis per barangay type to detect systematic under-prediction patterns flagged for coordinator override.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

Phase	Activity / Task	Tools Used	Expected Outcome
Sprint 1 (May 1-15)	Data ingestion: Project NOAH hazard CSV, HDX indicators, PSA 2020 CPH, OpenRouteService into Supabase PostGIS and pgRouting. Schema design, role-based auth, repository scaffolding.	GitHub, Python, GDAL, Supabase CLI	All barangay features are loaded and queryable via PostGIS. pgRouting returns valid shortest-path results between two barangay centroids. Next.js 16 renders separate coordinator and volunteer views behind authenticated routes.
Sprint 2 (May 16-31)	Core AI pipeline: TabPFN v2 FastAPI service. Silent Area scoring engine in Supabase. OR-Tools integrated with pgRouting. SEA-LION API parsing pipeline with confidence-threshold flagging.	VS Code, Python, FastAPI, OR-Tools, Supabase Edge Functions	Silent Area scores update correctly on a 15-minute cycle. TabPFN v2 returns supply quantity estimates for a test barangay set within 2 seconds. OR-Tools generates valid multi-team routes for 3 teams across 10 barangays in under 500ms. SEA-LION correctly extracts location, needs, and road status from 8 out of 10 sample Filipino or Bisaya test inputs.
Sprint 3 (June 1-15)	MapLibre operational map with three-state pins. Per-team route rendering. Offline PWA volunteer form with Workbox background sync and conflict resolution. Semaphore PH SMS webhook. Supabase	Next.js 16, MapLibre GL JS, Workbox, Semaphore PH	Coordinator map renders all barangay pins with correct state. A test SMS shortcode submission produces a map pin within 30 seconds. Volunteer PWA submits a queued offline report successfully upon reconnection with no data loss.

	Realtime volunteer tracking.		Realtime volunteer GPS positions update on coordinator map within 3 seconds of submission.
Sprint 4 (June 16-25)	System testing, TABPFNV2 MAE validation against HDX records, demo scenario preparation on Cebu 2025 earthquake data, pitch recording, production deployment. End-to-end testing, TabPFN v2 MAE validation against HDX historical records, full demo scenario on Cebu 2025 earthquake barangay data, pitch recording, production deployment.	Playwright, Vercel, Huggingface Spaces, OBS Studio	Playwright end-to-end tests pass for coordinator dispatch flow and volunteer report submission flow. TabPFN v2 MAE on held-out HDX supply records documented and lower than equal-distribution baseline. Full demo scenario runs without errors on real Cebu barangay data. System live on Vercel and Render.com with zero-cost configuration confirmed.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

Scaling Potential: LUWAS is geography-agnostic by architecture. The sole country-specific dependency is Semaphore PH, replaceable with any ASEAN-native SMS aggregator via a single configuration change. Scaling to a new country requires three steps: import OSM administrative boundaries, substitute Project NOAH with the equivalent national open hazard dataset (BNPB for Indonesia, DMH for Myanmar, DDPM for Thailand), and configure the local SMS gateway. Every AI model, routing engine, database schema, and frontend component is unchanged.

SEA-LION's training corpus natively covers Filipino, Bahasa Indonesia, Thai, Vietnamese, and Malay, making the NLP parsing pipeline ASEAN-ready without retraining. TabPFN v2's in-context learning architecture enables supply forecasting in any new country deployment using OCHA historical records as context, with zero locally labeled data required at launch. As real-world deployment data accumulates, the TabPFN v2 assistive layer provides a clear upgrade path: labeled field report outcomes feed a domain-specific fine-tuned regression model, progressively replacing in-context inference with a trained humanitarian logistics forecasting model purpose-built for ASEAN disaster contexts.

Infrastructure Cost: At pilot and demo scale, LUWAS operates at near-zero infrastructure cost: Vercel free tier for Next.js deployment, Supabase free tier for database and Realtime, and Hugging Face Spaces for FastAPI inference. Production deployments serving multiple simultaneous NGOs during an active disaster will incur scaling costs as Realtime connections and inference requests exceed free tier thresholds. This is documented transparently and addressed through the government SLA licensing tier of the business model, which funds production infrastructure.

Technical Constraints: TabPFN v2 forecast confidence degrades gracefully to population-weighted heuristics in zero-data deployments, disclosed via coordinator UI confidence indicators. pgRouting accuracy is bounded by OSM road network completeness in remote areas; low-confidence road segments are visually flagged on the operational map. SEA-LION API unavailability is fully mitigated by a Gemini 2.5 Flash structured extraction fallback. Offline-first synchronization via Workbox background sync introduces conflict resolution complexity in partial-connectivity scenarios.